

Introduction

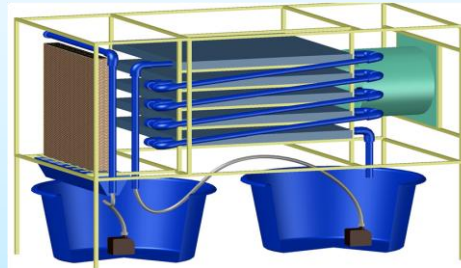
- ❖ Hot, dry regions face increasing cooling demand from rising temperatures.
- ❖ Conventional air conditioners consume lots of electricity and are expensive to run.
- ❖ Energy-efficient, eco-friendly alternatives lower bills and reduce demand.
- ❖ Evaporative coolers use water evaporation to efficiently lower air temperatures.
- ❖ Two-stage systems pair indirect cooling with direct wet pads.

Purposes of Evaporative Air Cooler

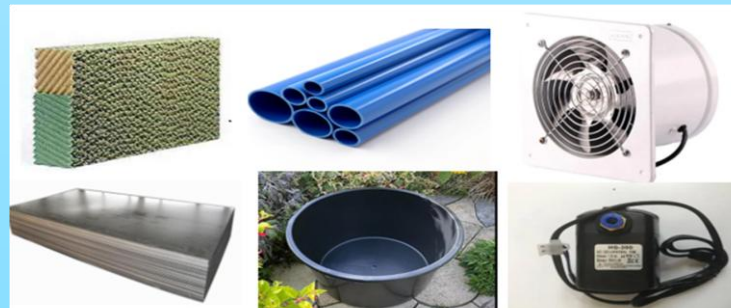
- ❖ Provide affordable cooling using evaporation, suitable for hot, dry climates.
- ❖ Reduce electricity consumption and operating costs compared with conventional ACs.
- ❖ Lower environmental impact by reducing emissions and overall energy demand.
- ❖ Improve indoor comfort by supplying cooler air while maintaining ventilation.
- ❖ Provide simple, low maintenance cooling systems adaptable to many applications.

Working Procedure of Evaporative Air Cooler

- ❖ Pump wets plates and pads
- ❖ Air sensibly cooled in flat plate heat exchanger
- ❖ Air evaporatively cooled through cellulose pads
- ❖ Cooled humid air deliver from pad



Components of Evaporative Air Cooler



**Technological University
(Mandalay)
Department of Mechanical
Engineering**

“Evaporative Air Cooler”



Types of Evaporative Air Cooler

- ❖ Direct Cooler- swamp coolers, more common in residential settings
- ❖ Indirect Cooler- used in industrial or high demand areas
- ❖ Direct, Indirect Cooler (Two Stage)- offers higher efficiency
- ❖ Hybrid Cooler- combine both mechanical refrigeration and evaporation

Fabrication Procedure

Materials

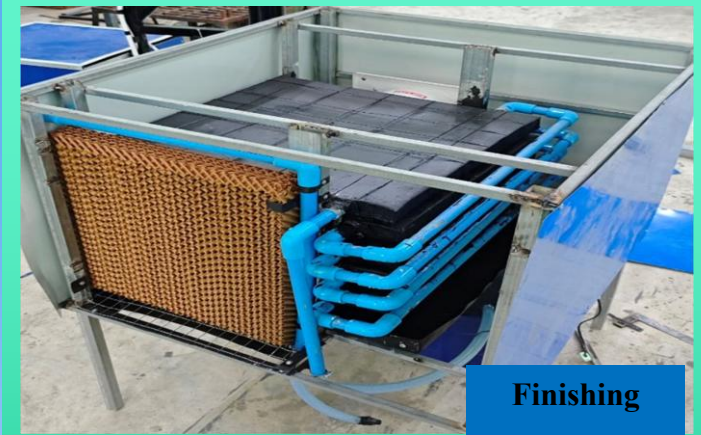
- ❖ Cooling Pad
- ❖ PVC Pipes
- ❖ Galvanized Iron Sheet
- ❖ Water Tank
- ❖ Submersible Water Pump
- ❖ Exhaust Fan
- ❖ Flat Plate Heat Exchanger
- ❖ Hollow Steel Frame

Methods

- ❖ Measuring
- ❖ Marking
- ❖ Cutting
- ❖ Welding
- ❖ Installation
- ❖ Finishing
- ❖ Testing



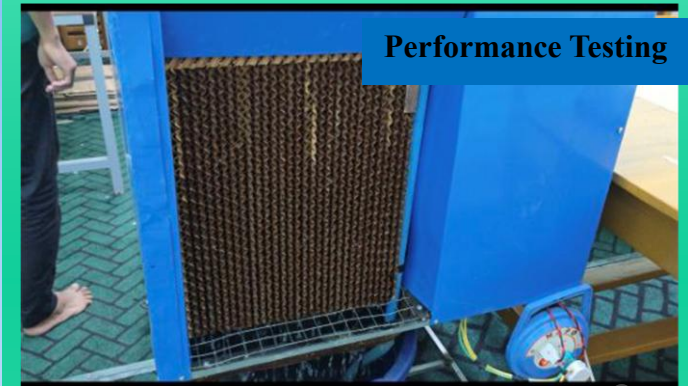
Cutting



Finishing



Welding



Performance Testing



Measuring

Marking



Installation

Performance Test Results

Time (min)	Outlet Air Temperature (°C)	Relative Humidity (%)
10:01 am	29	60.2
10:02 am	25.5	70.2
10:03 am	26	74.5
10:04 am	24.8	76.7
10:05 am	23.8	80